SChain: paper review and implementation upgrade

8 September 2026 · Updated code version 0.2.0

Reviewed inputs: Schain-mirza1(1).zip and all 12 distinct supplied papers. The separately uploaded main article and its project-source copy are byte-identical. The ZIP contains source code and tests, but no completed extraction dataset to assess as a real supply graph.

Recommendation

Improve the reliability of each edge before trying to grow a larger graph. Your code already implements much of the useful architecture: a static ontology, deterministic normalization, LLM entity resolution, node-centric upstream expansion, persistent state, source metadata, evidence checking and idempotent graph writes. These foundations should be retained. The strongest additions are stricter source grounding, preserved evidence, reliable chunk-level resume, correct graph tiers and an inspectable graph view.

This release implements those changes and adds an exact relation evaluator. It does not claim a new research method or measured extraction-accuracy improvement. That claim needs a controlled held-out comparison using your actual model and source corpus.

What differs from the main article?

The main article is AlMahri, Xu and Brintrup, Enhancing supply chain visibility with knowledge graphs and large language models (online 2025), DOI. Page numbers below refer to the supplied PDF including its publisher cover.

Point	Paper	Supplied code / retained choice
Ontology	Six entity types and four relations in quantitative evaluation	Same Company, Location, Material, Person, Product, Mine; suppliesTo, locatedIn, owns, produces
Extraction model	GPT-4o evaluation	Configurable local/API model; inherited Qwen setting is not a verified equivalent
Prompting	Described as zero-shot; definitions and entity examples	Includes a worked input/output demonstration, so not an exact zero-shot replication
Entity resolution	Second prompt groups equivalent names by type	Deterministic normalization, aliases, cache, then optional LLM grouping
Final case-study graph	At least three distinct credible sources per triple	One explicit validated source can create an edge; second-source search intentionally disabled

Point	Paper	Supplied code / retained choice
Source reputation	Credibility screening in case-study pipeline	Descriptive metadata; not an edge-acceptance threshold or confidence probability
Scope	Static explicit graph; case study additionally includes contains	Static four-relation ontology; configurable tier-limited upstream search
Focal company	Multiple EV firms in the case study	Configurable seeds; BMW is one possible seed, not the universal target

The paper's numerical entity-disambiguation score is a cluster identifier, **not a confidence score** (Figure 4, PDF p. 12). The existing single-source policy is a deliberate coverage-versus-corroboration choice, and this upgrade preserves it.

The paper reports all-relation F1 **0.328 ± 0.026** and suppliesTo precision **0.326 ± 0.170**, with suppliesTo recall **0.629 ± 0.183** (Table 1, PDF p. 16). These results make false supplier edges a serious concern: one mistaken edge can expand an entire irrelevant upstream branch. The reported entity-disambiguation precision@1 of 0.973 is a separate task and cannot establish graph correctness (Table 2, PDF p. 17).

Ideas implemented in the updated code

Rank	Concrete problem found	Implemented change	Expected benefit / boundary
1	document_text was passed into validation but not used; invented quotes could pass an LLM judge	Require the quoted evidence to occur in the actual document after Unicode/whitespace normalization	Rejects invented source wording before another model call; does not prove the quoted statement is true
2	Validation with no LLM manufactured a supports result	Keep deterministic-only evidence explicitly unchecked	Stops false claims of model validation; such evidence cannot satisfy the existing validated frontier gate
3	A retry could repeat all successful chunks	Atomic per-chunk checkpoints, stable chunk IDs, bounded transient retries and prompt/model-sensitive partial caches	Avoids repeating completed work after partial failure; real runtime gain depends on failure rate and document size
4	IGNORE could silently mark a partially extracted document complete	Default RAISE in both operating modes	Failure remains visible/retryable; explicit IGNORE remains an opt-in partial-result policy
5	Same-document triples collapsed different quotes/chunks; Neo4j replaced the latest quote	Evidence-specific fact IDs and additive quote/document/fact records on one semantic edge	Preserves every retained witness without inflating supplier-edge or document counts
6	Export quality flags were hard-coded true; cached support could be stale	Derive nullable validation flags and counts from the actual fact rows	Legacy unknowns remain unknown; source count and evidence count stay distinct

Rank	Concrete problem found	Implemented change	Expected benefit / boundary
7	Stale frontier pointers and alias variants could produce bad tiers or duplicate seeds	Alias-aware seed matching and indexed, bounded upstream BFS with supported parent paths	Correct shortest observed tiers, terminal boundary, and no phantom supplier paths
8	Resumed writes lost aliases; interrupted source persistence could leave missing provenance	Restore persisted entity resolution and source records; atomic JSON stage checkpoints	More faithful recovery without repeating successful extraction
9	Fresh-checkout service checks guessed endpoints or treated TCP as Neo4j	Read actual config with available Python; require Neo4j protocol/driver health	More reliable Linux server startup and ownership checks
10	JSON output was difficult to inspect and demonstrate	Self-contained offline HTML explorer and SVG export	Useful overview plus exact source inspection; no extra frontend package or CDN
11	External dataset overlap cannot measure extraction precision	Exact document-level directed triple evaluator, including negative documents	Enables a valid held-out quality comparison when independently annotated labels are supplied

The existing static ontology, INGEST/EXPAND modes, one-source policy, retrieval providers, LLM provider abstraction, seeds and Neo4j canonical graph remain. The new viewer is the explicit scope extension authorized by your request for a useful graph representation.

Key files: schain/kg/validate.py, schain/kg/extract.py, schain/kg/facts.py, schain/kg/writer.py, schain/pipeline.py, schain/graph_export.py, schain/checkpoint.py, schain/viewer.py, schain/viewer_assets/, schain/validation/gold.py, schain/__main__.py, and the existing server scripts.

The use of RAISE follows the documented chunk error behavior of the official Neo4j KG builder. The writer retains indexed uniqueness and idempotent MERGE, consistent with the official Cypher MERGE guidance. The installed API used for testing was neo4j-graphrag 1.19.0; no untested major API migration was introduced.

How the graph should be interpreted

The overview shows supplier-to-customer arrows and the shortest observed supplier tier relative to the configured seed set. Ownership, location and product relationships remain available as contextual layers, but do not create supply tiers. A seed need not be an OEM: all configured seed companies start at tier 0. The viewer does not reinterpret that tier as a business classification.

Select a relationship to read its exact quotes and source metadata. A source-matched quote and a model-supported relation are separate properties. Several documents can repeat one syndicated claim; support_count is not independent confirmation. Source reputation is not a calibrated probability. No observed path means only no path in the available graph, not absence of the real dependency. Extraction timestamps do not establish contractual start/end dates or whether an older relationship is still active.

The viewer has search/alias matching, per-seed upstream filtering, relation/type/tier/evidence filters, neighborhood focus, pan/zoom, a paginated relationship table, SVG download and filtered JSON

export. It reads local files and needs no internet or model. Its bounded SVG drawing intentionally displays at most 300 nodes and 1,000 edges, with a visible limit message; larger filtered datasets remain available in the table/JSON. The supplied demo uses fictional companies and manually authored source sentences, not extraction results.

What each supplied paper contributes

These are design transfers, not claims that this release reproduces each paper's system. The detailed reviews under docs/literature/ include full bibliographic identities, PDF page references, method summaries, results and limitations.

Source	Useful transfer to this implementation	Boundary
P01 AlMahri et al. (2025), Enhancing supply chain visibility...	Controlled schema, typed entity resolution, evidence and supplier direction (PDF pp. 8-17)	Model, prompting and three-source case-study gate differ
P02 Huang et al. (2025), ORLM, arXiv:2405.17743v5	Separate executable/schema validity from semantic correctness; hard negatives (pp. 15-19, 30)	Optimization-model training results are not KG extraction results
P03 Callefi et al. (2026), Building a pathway...	Establish structured data and semantic capabilities before higher autonomy (pp. 9-12, 16-19)	Capability hierarchy is not a measured pipeline speedup
P04 Song et al. (2026), LLMs in SCM: systematic review...	Source-linked supplier knowledge, provenance and contradiction awareness (pp. 21-23)	Conceptual review, not a new validated extraction algorithm
P05 Congès et al., R-IO SUITE, journal 2026 / HAL 2025	Deterministic metamodel checks and separation of persistent facts from proposals (pp. 22, 26-28)	Do not instantiate imagined roles as real supplier edges
P06 AlMahri et al. (2026), Automating Supply Chain Disruption Monitoring, arXiv:2601.09680	Deterministic graph traversal, JSON contracts, cached tools and tier views (pp. 9-18)	Assumes an existing KG; risk formulas are inconsistent across its body/case study
P07 Sukharamwala & Won (2026), LLM4SCM	Structured backend calculations remain authoritative; inspectable outputs (pp. 9, 14-16)	Solver/data integration requires quantities absent from this graph
P08 Quan et al. (2025), Risk Assessment in Hyperconnected Logistic Hub Network Deployment, arXiv:2503.21115	Bounded tool workflows and reuse of retrieved information (pp. 3-5)	Logistics-hub proof of concept, not supplier extraction benchmark
P09 Quan et al. (2026), multi-agent risk assessment / MARS	Source-constrained feedback, explicit rubrics and bounded revision (pp. 8-14)	Weather severity results do not validate supplier relations
P10 Trifone et al. (2026), Drivers and constraints of agentic AI...	Prioritize data quality, analyst oversight and understandable evidence (pp. 16-19, 32-34)	Nineteen-interview exploratory study, not performance measurements

Source	Useful transfer to this implementation	Boundary
P11 Baharmand & Behdani (2026), eco-friendly humanitarian supply chains	Trace numerical inputs to sources and keep scenario/proxy data separate (pp. 9-15)	Do not invent costs/capacities/emissions for a supplier graph
P12 Jannelli et al. (2025), Agentic LLMs...consensus-seeking	Tool/model combinations need ablation; extra coordination is not automatically better (pp. 11-17)	Inventory simulation requires demand, lead times and stock states absent here

Research ideas to consider next - not implemented

1. **Selective semantic review.** Recheck only ambiguous directional evidence and disputed entity merges. Compare one validator with one bounded independent revision. Do not assume adding agents improves quality enough to justify its latency and cost.

2. **Publisher/content-aware corroboration as a separate view.** Restore independent-source analysis only if you explicitly choose to change the current acceptance policy. First deduplicate syndicated text and distinguish primary announcements from repeated reporting.

3. **Corpus-level efficiency.** Benchmark Parquet reads/writes and global edge-support refresh before replacing them with partitioned tables or incremental updates. The current pipeline still rewrites shared Parquet artifacts and is intended for one writer; chunk caching does not solve that scaling limit.

4. **Temporal validity as a later schema/version.** Distinguish announced, planned, active and terminated contracts only with explicit text/time evidence and dedicated evaluation. This release remains static; file dates are not relationship validity intervals.

5. **Grounded question answering after graph quality is established.** Return deterministic paths and exact source excerpts for questions such as who supplies BMW within two tiers. Any prose explanation must reference those paths and must not invent numbers or certainty.

I do not recommend adding autonomous procurement, inventory negotiation, supplier replacement, or numerical risk/optimization scores now. The supplied papers require operational variables that this code does not contain. A large or visually dense graph is not evidence of a better supply-chain model.

How to demonstrate an actual improvement

Use the same held-out documents, model settings, retrieval budget, ontology and seeds for the original and upgraded pipelines. Independently label the correct canonical directed triples, including documents with no relation, reversed roles, subsidiaries, vague partnerships, planned deals, repeated news and multi-chunk articles. Keep prompt examples separate.

Report directed relation precision/recall/F1 overall and per relation; supplier-edge precision is especially important because expansion amplifies mistakes. Separately record false entity merges, source-quote grounding failures, accepted/rejected facts, supported reachable company count, wall-clock time, actual model calls/tokens, retries and interrupted-run recovery. An ablation should isolate grounding, caching and any later review stage. Higher source/edge counts alone are not a

success metric. The exact evaluator provided here scores accepted triples; NER and entity-resolution quality need their own labels and metrics.

Validation and practical limits

The final test suite completed with **421 passed, 4 skipped** (86.39 seconds; six existing pandas FutureWarnings). See TEST_RESULTS.md for the environment. Checks exercise parsing, source grounding, negative documents, directed edge scoring, cache failure/recovery, metadata, Neo4j query parameters, tiers, seeds and shell probes. Four existing integration tests require external services and are skipped when unavailable. Python tests use model/database stubs; JavaScript syntax and offline packaging are checked. The actual SVG canvas was also rendered and visually inspected with Node/CairoSVG; the synthetic default view has nine entities and eight supply edges. No live browser interaction, model endpoint, retrieval experiment, GPU inference, or Neo4j server integration was run.

Legacy document checkpoints are intentionally reused. New source validation is not retroactive. Use separate storage and a separate Neo4j database/instance for a clean comparison. The --restart flag deletes configured local generated artifacts but leaves Neo4j intact; it cannot remove stale database edges. New IDs preserve quote/chunk evidence going forward; previously lost quotes need fresh extraction. A crash after a model response but before its checkpoint may still repeat that call. JSON checkpoints are atomic, while existing Parquet storage remains single-writer and is not transactional across all layers.

Follow START_HERE.md for installation and commands. This release makes the implementation more auditable and resilient; measured accuracy and performance gains remain to be established.